# Highlights of Linear Algebra

## Vectors, Matrices, and Geometric Intuition

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MAI 600: Mathematics and Statistics for AI

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# Why is Linear Algebra the Bedrock of AI?

## 1. It's the Language of Data

Linear algebra gives us a universal way to represent and work with data.

- **Vectors** represent individual data points (a user, a word, a patient).
- **Matrices** represent entire datasets or systems (a collection of users, an image, the weights of a neural network).

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## 2. It's the Engine of Computation

Many AI algorithms are fundamentally a series of linear algebra operations.

- **Deep Learning:** A neural network's forward pass is just a sequence of matrix multiplications and vector additions.
- **Dimensionality Reduction (PCA):** Finds the most important "directions" in the data.

## Key AI Applications

- **Computer Vision:** Images are matrices of pixels. Rotations, scaling, and filtering are all matrix operations (https://www.youtube.com/watch?v=aircAruvnKk).
- **Natural Language Processing (NLP):** Word embeddings like Word2Vec represent words as vectors, capturing semantic meaning.
- **Recommender Systems:** Analyze user-item matrices to predict what you might like next (e.g., Netflix, Amazon).
  https://www.youtube.com/watch?v=aircAruvnKk

# Lecture Outline


1. Vectors: The Data Atoms
2. Vector Spaces, Norms, and Inner Products
3. Matrices and Linear Transformations

A vector is an object with both **magnitude** (length) and **direction**. We can think of it in two ways:

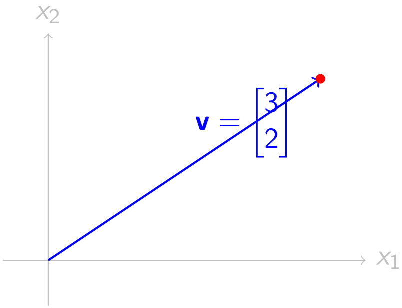
## Algebraically

An ordered list of numbers. In AI, we typically write vectors as **column vectors**.

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} 0.8 \\ 1.2 \\ -3.5 \\ 0.1 \end{bmatrix}$$

## Geometrically

An arrow in space pointing from the origin to a coordinate point.



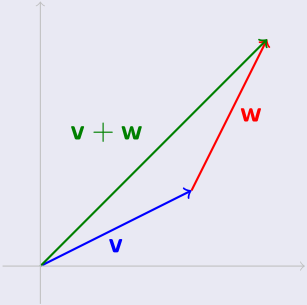

## Example: A Feature Vector

A vector can represent a single data point. For a housing dataset:

$$\text{house} = \begin{bmatrix} \text{Area (sq ft)} \\ \text{Bedrooms} \\ \text{Age (years)} \end{bmatrix} = \begin{bmatrix} 1500 \\ 3 \\ 10 \end{bmatrix}$$

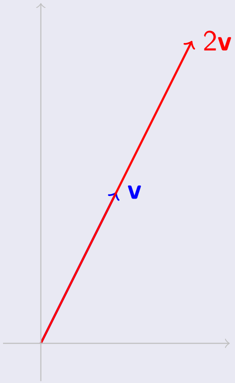
## Vector Addition (Tip-to-Tail Rule)

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a+c \\ b+d \end{bmatrix}$$



# Scalar Multiplication (Stretching/Shrinking)

$$c \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ca \\ cb \end{bmatrix}$$



# Practice Session: Drawing and Adding Vectors

## Problem

Given the vectors $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$.

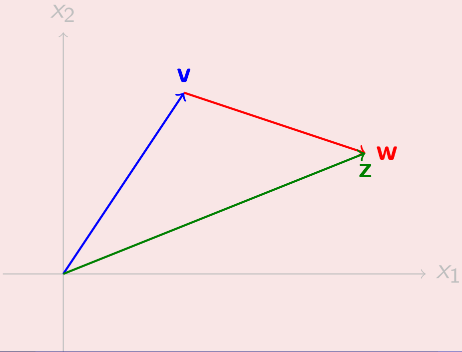
1. Calculate the resultant vector $\mathbf{z} = \mathbf{v} + \mathbf{w}$.
2. On a 2D grid, draw $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{z}$. Show the tip-to-tail addition graphically.

## Solution

1. **Calculation:**

$$\mathbf{z} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2+3 \\ 3-1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

2. **Drawing:**



# Vector Spaces

**Definition:** A set $V$ closed under addition and scalar multiplication.
**Example:** $\mathbb{R}^2$ with vectors $(x, y)$.
- $u = (1, 2), v = (3, 4)$ - $u + v = (4, 6)$ - $2u = (2, 4)$
This stays inside $\mathbb{R}^2$, so it's a vector space.

# Subspaces and Span

**Span:** All linear combinations of vectors.
**Example:** $\text{Span}\{(1,0),(0,1)\} = \mathbb{R}^2$.
**AI Link:** Features in ML can be represented as spanning vectors in a high-dimensional space.

### Example: Word Embedding Space

In NLP, we don't have just 2 basis vectors, but perhaps 300! Each basis vector might represent some abstract linguistic feature. A word like "learning" is then defined by its unique combination of these 300 basis vectors. The entire vocabulary lives within this 300-dimensional vector space.

# Vector Norms: Measuring Length

A **norm** is a function that measures the length or magnitude of a vector. It's a key tool for measuring error or distance.

## L2 Norm (Euclidean Distance)

The most common "straight-line" distance.

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Used everywhere, from calculating the error of a prediction to finding nearest neighbors.

## L1 Norm (Manhattan Distance)

The "city block" distance; sum of absolute components.

$$\|\mathbf{v}\|_1 = |v_1| + |v_2| + \cdots + |v_n|$$

Often used in machine learning when we want to encourage sparsity (many zero values).

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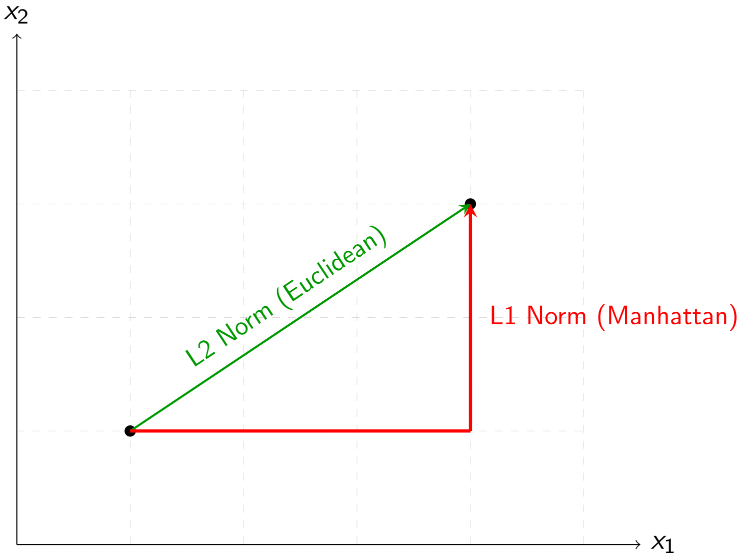
$x_2$
L2 Norm (Euclidean)
L1 Norm (Manhattan)
$x_1$

# Practice Session: Calculating Norms

## Problem

Imagine a simple feature vector for a movie recommendation system is $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$,

representing scores for action, comedy, and drama.

1. Calculate the L2 norm (Euclidean length) of the vector, $\|\mathbf{v}\|_2$.
2. Calculate the L1 norm (Manhattan length) of the vector, $\|\mathbf{v}\|_1$.

## Solution

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2. **L1 Norm:**

$$\|\mathbf{v}\|_1 = |2| + |-3| + |6| = 11$$

The **inner product** (or **dot product**) takes two vectors and returns a single scalar.

$$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n$$

**Example**

Let $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ and $= \begin{bmatrix} 5 \\ 4 \\ 2 \end{bmatrix}$. Then the dot product is $\mathbf{v} \cdot \mathbf{w} = 10 + 12 + (-2) = 20$.

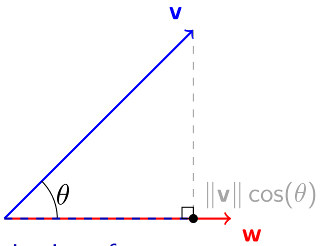
# The Inner Product: Projection and Similarity

Its real power comes from its geometric interpretation:

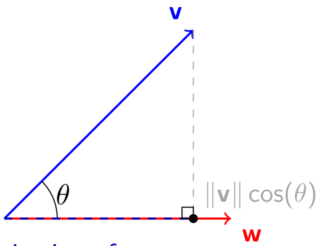
$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\|_2 \|\mathbf{w}\|_2 \cos(\theta)$$

The inner product (or dot product) $\mathbf{v} \cdot \mathbf{w}$ tells us about the **alignment** of two vectors. Geometrically, it's related to how much of one vector "points in the direction of" or "projects onto" the other.

# Inner Product: Geometric Intuition (Projection)




Projection of $\mathbf{v}$ onto $\mathbf{w}$




## Key Takeaway

The quantity $\|\mathbf{v}\| \cos(\theta)$ is the **length** of the projection of **v** onto **w**. The dot product is simply this length multiplied by the length of **w**.

- If **v** and **w** point in the same direction ($\theta = 0^\circ$), $\cos(\theta) = 1$, dot product is maximal.
- If they are perpendicular ($\theta = 90^\circ$), $\cos(\theta) = 0$, dot product is zero.

## Application: Cosine Similarity

By rearranging the formula, we get a direct measure of similarity that ignores vector magnitudes:

$$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|_2 \|\mathbf{w}\|_2}$$

This is the engine behind recommender systems ("users who liked X also liked Y") and document search. It measures the similarity of direction of user preference vectors or document topic vectors.

## Problem

Let's model two users' movie ratings. User A's ratings for (Star Wars, Titanic) are $\mathbf{a} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$.

User B's ratings are $\mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

1. Calculate the inner product $\mathbf{a} \cdot \mathbf{b}$.
2. Calculate the cosine similarity between $\mathbf{a}$ and $\mathbf{b}$. Do these users have similar taste?

## Solution

1. **Inner Product:**

$$\mathbf{a} \cdot \mathbf{b} = (5)(4) + (1)(2) = 22$$

Since the result is a large positive number, we know the vectors are generally pointing in the same direction.

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2. **Cosine Similarity:** First, we need the norms.
   - $\|\mathbf{a}\|_2 = \sqrt{5^2 + 1^2} = \sqrt{26}$
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   Now, apply the formula:

$$\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} = \frac{22}{\sqrt{26}\sqrt{20}} \approx 0.96$$

   A cosine similarity of 0.96 is very close to 1, indicating that the users have very similar taste profiles.

A matrix is a rectangular grid of numbers. In AI, they have two primary roles:

## 1. As containers for data

- An entire dataset, where rows are samples and columns are features.
- A grayscale image, where each element is a pixel intensity.
- A batch of data fed into a neural network.

$$\text{Data} = \begin{bmatrix} \text{— house 1 —} \\ \text{— house 2 —} \\ \vdots \\ \text{— house m —} \end{bmatrix}$$

## 2. As linear transformations

A matrix can be seen as a function that transforms vectors from one space to another.

- The weights of a neural network layer form a matrix that transforms the input data.
- Matrices can rotate, scale, and shear vectors.

$$\mathbf{y} = A\mathbf{x} \quad \text{(Matrix A "acts on" vector x)}$$

## Addition and Scalar Multiplication

Just like vectors, matrices can be added (if they have the same dimensions) and multiplied by scalars. This is done element-wise.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix} \qquad k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

## Problem

Given two matrices $A$ and $B$, which might represent two different sets of features for the same batch of data:

$$A = \begin{bmatrix} 1 & 5 \\ 3 & 0 \\ -2 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 2 \\ -1 & 7 \\ 5 & -3 \end{bmatrix}$$

Calculate the sum $A + B$.

## Solution

Matrix addition is performed element-wise:

$$
\begin{aligned}
A + B &= \begin{bmatrix} 1 & 5 \\ 3 & 0 \\ -2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -1 & 7 \\ 5 & -3 \end{bmatrix} \\
&= \begin{bmatrix} 1+0 & 5+2 \\ 3+(-1) & 0+7 \\ -2+5 & 4+(-3) \end{bmatrix} \\
&= \begin{bmatrix} 1 & 7 \\ 2 & 7 \\ 3 & 1 \end{bmatrix}
\end{aligned}
$$

The resulting matrix has the same dimensions as $A$ and $B$.

## The Most Important Operation: Matrix Multiplication

Matrix multiplication is how transformations are applied and combined.

**Matrix-Vector Multiplication:** Transforms a vector.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

The result is a linear combination of the columns of the matrix.

**Matrix-Matrix Multiplication:** Represents a composition of transformations. If $C = AB$, applying matrix $C$ is the same as applying $B$ first, then $A$.

## Problem

Suppose a neural network layer has a weight matrix $W$ and receives an input vector $\mathbf{x}$:

$$W = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{bmatrix} \qquad \mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix}$$

Calculate the output vector $\mathbf{y} = W\mathbf{x}$ before the bias and activation function are applied.

## Solution

We multiply each row of the matrix by the column vector:

$$\mathbf{y} = W\mathbf{x} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix}$$
$$= \begin{bmatrix} (2)(4) + (-1)(1) + (0)(5) \\ (1)(4) + (3)(1) + (-2)(5) \end{bmatrix}$$
$$= \begin{bmatrix} 7 \\ -3 \end{bmatrix}$$

The input vector in $\mathbb{R}^3$ was transformed into an output vector in $\mathbb{R}^2$.

## Example: Matrix-Matrix Multiplication

### Problem

Given two matrices $A$ and $B$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Calculate the product $C = AB$.

## Solution

To calculate $C = AB$, we multiply rows of $A$ by columns of $B$:

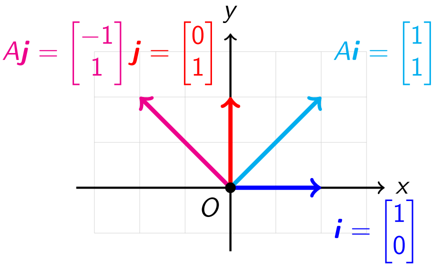
$$\begin{aligned} C &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \\ &= \begin{bmatrix} (1)(5) + (2)(7) & (1)(6) + (2)(8) \\ (3)(5) + (4)(7) & (3)(6) + (4)(8) \end{bmatrix} \\ &= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \end{aligned}$$

Remember, matrix multiplication is generally not commutative ($AB \neq BA$).

A matrix transforms the entire vector space. The key insight is to see what it does to the basis vectors **i** and **j**. The columns of a matrix tell you where the basis vectors land after the transformation.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

This matrix sends $\mathbf{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.



# Practice Session: Applying a Transformation

## Problem

Consider the "shear" transformation matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and the vector $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

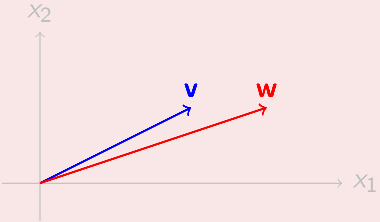
1. Calculate the transformed vector $\mathbf{w} = A\mathbf{v}$.
2. Draw $\mathbf{v}$ and $\mathbf{w}$ on a 2D grid. What is the geometric effect of this transformation?

## Solution

**1. Calculation:**

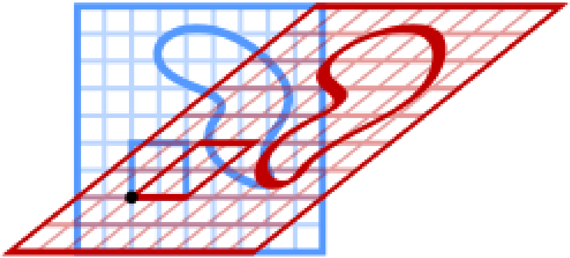
$$\mathbf{w} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} (1)(2) + (1)(1) \\ (0)(2) + (1)(1) \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

**2. Geometric Effect:**




The transformation shifted the top of the vector to the right. It "sheared" the space horizontally.

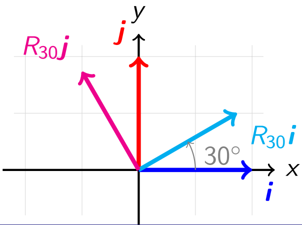
# Shear mapping



A rotation is a linear transformation. We can see what it does by observing its effect on the basis vectors $\boldsymbol{i}$ and $\boldsymbol{j}$.
The columns of the rotation matrix tell you where the basis vectors land. For a counter-clockwise rotation by 30°, the matrix is:

$$R_{30} = \begin{bmatrix} \cos(30^\circ) & \cos(120^\circ) \\ \sin(30^\circ) & \sin(120^\circ) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

This matrix sends $\boldsymbol{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$ and $\boldsymbol{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$.



# Summary

- **Vectors** are the atoms of our data, representing single data points with magnitude and direction.
- **Matrices** are collections of data or, more powerfully, **linear transformations** that manipulate vectors.
- **Vector Spaces** provide the "universe" where our data lives, and their properties are defined by basis vectors.
- **Norms** measure size/distance, while **Inner Products** measure similarity/projection. These are critical for loss functions and similarity metrics.
- **Matrix Multiplication** is the engine of AI, from applying transformations in computer vision to processing data in neural networks.

Questions?